UIL COMPUTER SCIENCE WRITTEN TEST

2026 REGION

APRIL 2026

General Directions (Please read carefully!)

1. DO NOT OPEN THE EXAM UNTIL TOLD TO DO SO.
2. There are 40 questions on this contest exam. You will have 45 minutes to complete this contest.
3. All answers must be legibly written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Clean erasures are necessary for accurate grading.
4. You may write on the test packet or any additional scratch paper provided by the contest director, but NOT on the answer sheet, which is reserved for answers only.
5. All questions have ONE and only ONE correct answer. There is a 2-point penalty for all incorrect answers.
6. Tests may not be turned in until 45 minutes have elapsed. If you finish the test before the end of the allotted time, remain at your seat and retain your test until told to do otherwise. You may use this time to check your answers.
7. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
8. All provided code segments are intended to be syntactically correct, unless otherwise stated. You may also assume that any undefined variables are defined as used.
9. A reference to many commonly used Java classes is provided with the test, and you may use this reference sheet during the contest. AFTER THE CONTEST BEGINS, you may detach the reference sheet from the test booklet if you wish.
10. Assume that any necessary import statements for standard Java SE packages and classes (e.g., `java.util`, `System`, etc.) are included in any programs or code segments that refer to methods from these classes and packages.
11. NO CALCULATORS of any kind may be used during this contest.

Scoring

1. Correct answers will receive **6 points**.
2. Incorrect answers will lose **2 points**.
3. Unanswered questions will neither receive nor lose any points.
4. In the event of a tie, the student with the highest percentage of attempted questions correct shall win the tie.

STANDARD CLASSES AND INTERFACES – SUPPLEMENTAL REFERENCE

package java.lang

```
class Object
  boolean equals(Object anotherObject)
  String toString()
  int hashCode()

interface Comparable<T>
  int compareTo(T anotherObject)
    Returns a value < 0 if this is less than anotherObject.
    Returns a value = 0 if this is equal to anotherObject.
    Returns a value > 0 if this is greater than anotherObject.

class Integer implements Comparable<Integer>
  Integer(int value)
  int intValue()
  boolean equals(Object anotherObject)
  String toString()
  String toString(int i, int radix)
  int compareTo(Integer anotherInteger)
  static int parseInt(String s)

class Double implements Comparable<Double>
  Double(double value)
  double doubleValue()
  boolean equals(Object anotherObject)
  String toString()
  int compareTo(Double anotherDouble)
  static double parseDouble(String s)

class String implements Comparable<String>
  int compareTo(String anotherString)
  boolean equals(Object anotherObject)
  int length()
  String substring(int begin)
    Returns substring(begin, length()).
  String substring(int begin, int end)
    Returns the substring from index begin through index (end - 1).
  int indexOf(String str)
    Returns the index within this string of the first occurrence of str. Returns
    -1 if str is not found.
  int indexOf(String str, int fromIndex)
    Returns the index within this string of the first occurrence of str, starting
    the search at fromIndex. Returns -1 if str is not found.
  int indexOf(int ch)
  int indexOf(int ch, int fromIndex)
  char charAt(int index)
  String toLowerCase()
  String toUpperCase()
  String[] split(String regex)
  boolean matches(String regex)
  String replaceAll(String regex, String str)

class Character
  static boolean isDigit(char ch)
  static boolean isLetter(char ch)
  static boolean isLetterOrDigit(char ch)
  static boolean isLowerCase(char ch)
  static boolean isUpperCase(char ch)
  static char toUpperCase(char ch)
  static char toLowerCase(char ch)

class Math
  static int abs(int a)
  static double abs(double a)
  static double pow(double base, double exponent)
  static double sqrt(double a)
  static double ceil(double a)
  static double floor(double a)
  static double min(double a, double b)
  static double max(double a, double b)
  static int min(int a, int b)
  static int max(int a, int b)
  static long round(double a)
  static double random()
    Returns a double greater than or equal to 0.0 and less than 1.0.
```

package java.util

```
interface List<E>
class ArrayList<E> implements List<E>
  boolean add(E item)
  int size()
  Iterator<E> iterator()
  ListIterator<E> listIterator()
  E get(int index)
  E set(int index, E item)
  void add(int index, E item)
  E remove(int index)

class LinkedList<E> implements List<E>, Queue<E>
  void addFirst(E item)
  void addLast(E item)
  E getFirst()
  E getLast()
  E removeFirst()
  E removeLast()

class Stack<E>
  boolean isEmpty()
  E peek()
  E pop()
  E push(E item)

interface Queue<E>
class PriorityQueue<E>
  boolean add(E item)
  boolean isEmpty()
  E peek()
  E remove()

interface Set<E>
class HashSet<E> implements Set<E>
class TreeSet<E> implements Set<E>
  boolean add(E item)
  boolean contains(Object item)
  boolean remove(Object item)
  int size()
  Iterator<E> iterator()
  boolean addAll(Collection<? extends E> c)
  boolean removeAll(Collection<?> c)
  boolean retainAll(Collection<?> c)

interface Map<K,V>
class HashMap<K,V> implements Map<K,V>
class TreeMap<K,V> implements Map<K,V>
  Object put(K key, V value)
  V get(Object key)
  boolean containsKey(Object key)
  int size()
  Set<K> keySet()
  Set<Map.Entry<K, V>> entrySet()

interface Iterator<E>
  boolean hasNext()
  E next()
  void remove()

interface ListIterator<E> extends Iterator<E>
  void add(E item)
  void set(E item)

class Scanner
  Scanner(InputStream source)
  Scanner(String str)
  boolean hasNext()
  boolean hasNextInt()
  boolean hasNextDouble()
  String next()
  int nextInt()
  double nextDouble()
  String nextLine()
  Scanner useDelimiter(String regex)
```

STANDARD CLASSES AND INTERFACES – SUPPLEMENTAL REFERENCE

Package java.util.function

```
Interface BiConsumer<T,U>
  void accept(T t, U u)

Interface BiFunction<T,U,R>
  R apply(T t, U u)

Interface BiPredicate<T,U>
  boolean test(T t, U u)

Interface Consumer<T>
  void accept(T t)

Interface Function<T,R>
  R apply(T t)

Interface Predicate<T>
  boolean test(T t)

Interface Supplier<T>
  T get()
```

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Note: Correct responses are based on **Java SE Development Kit 25 (JDK 25)** from Oracle, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g., "error" is an answer choice) and any necessary Java SE 25 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported using: `import static java.lang.System.*;`**

Question 1

What base would 1212 need to be in to ensure that when converted to base 10, the value is between 60_{10} and 150_{10}?

A) Base 3 **B)** Base 4 **C)** Base 5 **D)** Base 6 **E)** Base 7

Question 2

What is output by the code to the right?

A) `5.5`

B) `6`

C) `5`

D) `6.0`

E) `4.0`

```
int a = 6;
double b = 3.0;
int c = 4;

out.println(a * c / b + 10 % c -
            (int)(b + 1.5));
```

Question 3

How many lines of output are generated by the code to the right?

Lines containing only a newline character do not count. The following code produced only one line of output for example.

```
out.println("Hello World");
```

A) 8 Lines **B)** 9 Lines

C) 10 Lines **D)** 11 Lines

E) 12 Lines

```
String s = "hi\nthere";
out.println(s);
out.printf("%s\n%s\n", "one", "two\nthree");
out.print("a\n\nb");
out.println("x\ny");
out.printf("end%s", "\n");
```

Question 4

What is output by the code to the right?

A) `WorldWorldHelloHello`

B) `HelloHelloHelloHello`

C) `HelloWorldHelloWorld`

D) `WorldHelloHelloWorld`

E) `HelloHelloWorldHello`

```
String Hello = "World";
String World = "Hello";
String Hello_World = World.repeat(2) +
            Hello + "Hello";

out.println(Hello_World);
```

Question 5

What is output by the code to the right?

A) `false`

B) `true`

```
boolean a = true;
boolean b = false;
boolean c = true;

boolean result = (a ^ b) &&
        !(b || c) || (a && c);
out.println(result);
```

Question 6

What is output by the code to the right?

A) `7 3.0`

B) `7 4.0`

C) `4 3.0`

D) `7 3.6`

```
int a = -7;
double b = 3.6;

out.println(
        Math.max(Math.abs(a), Math.round(b))
        + " " + Math.min(
            Math.floor(b), Math.ceil(b)
        ));
```

Question 7

What is output by the code to the right?

A) `5`

B) `6`

C) `3`

D) `7`

E) `13`

```
int a = 6;
int b = 3;
char c = 'C';
boolean flag = (a & b) > 1;

int result = (c & 0x0F) |
        (flag ? a ^ b : a & b);
out.println(result);
```

Question 8

What is output by the code to the right?

A) `501 99`

B) `500 100`

C) `100`

D) `501`

E) `501 100`

```
int x = 100;
int y = 500;

if (x > 'a' | y++ < 501) {
    out.println(y);
}
if (-x-- > ~x) {
    out.println(x);
}
```

Question 9

What is output by the code to the right?

A) `10`

B) `12`

C) `0`

D) There is no output due to a runtime error.

E) There is no output due to a compile time error.

```
for (int i = -1, total = 0; i < 19; i+=3) {
    int x = i / (i % 7);
    total += i;
}
out.println(total);
```

Question 10

What is output by the code to the right?

A) `[2, 5, 3, 4, 3]`

B) `[1, 3, 3, 4, 3]`

C) `[2, 3, 3, 4, 5]`

D) `[1, 5, 3, 4, 5]`

E) There is no output due to an error.

```
int [] arr = new int[] {1, 2, 3, 4, 5};
arr[0] = arr[arr[0]];
arr[1] += arr[4] = arr[2];
out.println(Arrays.toString(arr));
```

Question 11

Which of the following interfaces does Scanner **not** implement?

A) `Closeable`

B) `AutoCloseable`

C) `Iterator`

D) `Object`

E) `Scanner` implements all of the above.

Question 12

Which of the following values is output on its own line by the code to the right?

A) 899 **B)** 18

C) 101 **D)** 50

E) 99

```
for (int i = 10; i >= 1; i--) {
    int product = i * (i - 1);
    for (int j = 1; j < i; j++) {
        product += product / j;
    }
    out.println(product);
}
```

Question 13

Which of the following is the correct order of precedence from highest to lowest?

A) ++(postfix), ~, *, ==, |(bitwise), &&

B) ~, ++(postfix), *, ==, |(bitwise), &&

C) ++(postfix), ~, *, |(bitwise), ==, &&

D) ++(postfix), *, ~, ==, |(bitwise), &&

E) *, ~, ++(postfix), ==, &&, |(bitwise)

Question 14

Which of the following is the correct order of floating-point values smallest to largest?

A) negative infinity, finite negative values, negative zero, zero, finite positive values, infinity

B) negative infinity, finite negative values, zero, negative zero, finite positive values, infinity

C) Both A and B.

Question 15

What is the largest possible sum of values in the list?

A) 4950 **B)** 5050

C) 2475 **D)** 2525

E) 0

```
List<Integer> list = new ArrayList<>();

for (int i = 0; i < 100; i++) {
    list.add(i);
    int value =
        (int)(Math.random() * list.size());
    Integer t = Integer.valueOf(value);
    list.remove(t);
}
```

Question 16

What is output by the code to the right?

A) 23 **B)** 13

C) There is no output due to a compile error.

D) There is output, followed by a runtime error.

E) There is no output due to a runtime error.

```
try {
   String[] st = new String[5];
   Object[] o = st;
   o[2] = 39.98;
   out.print(1);
} catch (ArrayStoreException e) {
   out.print(2);
} finally {
   out.print(3);
}
```

Question 17

What could replace **<1*>** in the code to the right so that it will compile and run without error (removing items).

A) `Queue<Integer>` **B)** `PriorityQueue<>`

C) `Queue<>` **D)** Either A or B. **E)** All of the above.

Question 18

Assuming **<1*>** is filled in correctly, what is output by the code to the right?

A) `[5, 6, 7, 8, 9, 10, 12]`

B) `[5, 6, 8, 7, 9, 12, 10]`

C) `[5, 6, 8, 9, 7, 12, 10]`

D) `[2, 3, 4, 5, 6, 9, 10]`

E) There is no output due to a runtime error.

Question 19

If the integer array `i` was rewritten to instead be of length N, what would be the runtime of the code to the right?

A) $\mathcal{O}(N \log N)$ **B)** $\mathcal{O}(N)$

C) $\mathcal{O}(\log N)$ **D)** $\mathcal{O}(N^2 \log N)$

E) $\mathcal{O}(N^2)$

```
<1*> pq;
pq = new PriorityQueue<>();
int[] i = new int[] {
    12, 8, 7, 3, 9, 5, 10, 2, 4, 6
};

for(int j = 0; j < i.length; j++)
   pq.add(i[j]);

pq.poll();
pq.poll();
pq.poll();
out.println(pq);
```

Question 20

Which of the lines in the code to the right first causes an error?

A) `//1` **B)** `//2` **C)** `//3` **D)** `//4`

E) None of the above lines cause any errors.

```
List a;                    //1
a = new ArrayList(10);     //2
a.add("Someone");          //3
a.add(a.size(), 43);       //4
```

Question 21

What is output by the line marked `//q21` in the code to the right?

A) `70` **B)** `0` **C)** `35` **D)** `140`

E) There is no output due to infinite recursion.

Question 22

What is output by the line marked `//q22` in the code to the right?

A) `280` **B)** `0` **C)** `7` **D)** `70`

E) There is no output due to infinite recursion.

```
public int recur(int i) {
    if(i % 7 == 1)
        return 7;
    return (i % 7) * recur(i + 3);
}

//////////client code//////////
out.println(recur(19));  //q21
out.println(recur(2));   //q22
```

Question 23

What can replace **<1*>** in the code to the right so that the code compiles and runs without error?

A) `implements Comparable<Num>`

B) `implements Comparable`

C) Nothing is required **D)** Either A or B.

E) Either B or C. **F)** Any of the above.

Question 24

Which of the following terms apply to the `Num` class to the right?

A) Overriding **B)** Overloading **C)** Encapsulation

D) Both A and B **E)** All of the above **F)** None of the above

Question 25

Which of the following labeled methods in `Num` is required to be defined so that the code compiles and runs without error?

A) `//1` **B)** `//2`

C) `//3` **D)** `//4`

E) `//5` **F)** `//6`

G) B, E, and F. **H)** B and E.

I) B, C, and E. **J)** A, B, C, D, and E.

K) B, C, E, and F **L)** All of the above.

Question 26

What can replace **<2*>** in the code to the right so that the `ts` structure orders the numbers in decreasing order?

A) Nothing is required

B) `Collections.reverseOrder()`

C) `-1`

D) Either A or C.

E) Any of the above.

Question 27

Assuming **<1*>** and **<2*>** are filled in correctly, and that no other changes have been made to the `Num` class, what is output by the code to the right?

A) `[4.5, 39.8, 212, 1234]`

B) `[1234, 212, 39.8, 4.5]`

C) `[null1234, null212, null39.8, null4.5]`

D) `[null4.5, null39.8, null212, null1234]`

E) There is no output due to a compile error.

F) There is no output due to a runtime error.

```
class Num <1*> {
    double val;
    String o;

    public Num(int n) { //1
        o += n;
        val = n;
    }

    public Num(double n) { //2
        o += n;
        val = n;
    }

    public Num(long n) { //3
        o += n;
        val = n;
    }

    public Num(float n) { //4
        o += n;
        val = n;
    }

    public int compareTo(Num n) { //5
        return Double.compare(val, n.val);
    }

    public String toString() { //6
        return o;
    }
}

//////////////client code///////////////
TreeSet<Num> ts = new TreeSet<Num>(<2*>);
ts.add(new Num(212L));
ts.add(new Num(1234));
ts.add(new Num(4.5f));
ts.add(new Num(39.8));
out.println(ts);
```

Question 28

Which of the following sorting algorithms is demonstrated in the code to the right?

A) Tim Sort **B)** Bubble Sort

C) Quick Sort **D)** Merge Sort

E) Insertion Sort

```
ArrayList<Integer> a;
a = new ArrayList<>();
for(int y = 0; y < 212; y++)
    a.add((int)(Math.random() * y * y));
Collections.sort(a);
out.println(a);
```

Question 29

Which of the following is **not** a legal Java identifier?

A) Ca$H **B)** Final **C)** `Hamb_rger` **D)** `Number1`

E) All of the above are legal identifiers.

Question 30

What is output by the line marked `//q30` in the client code to the right?

A) `false false` **B)** `false true`

C) `true false` **D)** `true true`

E) There is no output due to a compile error.

Question 31

What is output by the client code to the right?

A) `false false` **B)** `false true`

C) `true false` **D)** `true true`

E) There is no output due to a compile error.

```
String s1 = "C3P0E5";
String s2 = "LikeStarWars";
String r = "([A-Z][0-9]?)+";
String o = "" + s1.matches(r);
o += " " + s2.matches(r);
out.println(o); //q30

r = "(([A-Z]\\w*)|(\\D\\W)){2,4}";
o = "" + s1.matches(r);
o += " " + s2.matches(r);
out.println(o); //q31
```

Question 32

What is output by the code to the right?

A) `[2, 3, 5, 7]`

B) `[7, 5, 3, 2]`

C) `[5, 7]`

D) `[3, 2]`

E) There is no output due to a runtime error.

```
List<Integer> nums = Arrays.asList(
        2, 3, 5, 7, 11, 13, 17, 19);

List<Integer> result = nums.stream()
        .filter(n -> n < 10)
        .sorted((a, b) -> b - a)
        .limit(4)
        .skip(2)
        .collect(Collectors.toList());

out.println(result);
```

Question 33

What is the Big-$\mathcal{O}$ time complexity of the `mystery` method to the right in terms of n?

A) $\mathcal{O}(n)$ **B)** $\mathcal{O}(n \log n)$

C) $\mathcal{O}(n^2)$ **D)** $\mathcal{O}(\log n)$

E) $\mathcal{O}(2^n)$

```
void mystery(int n) {
    for (int i = 0; i < n; i += 2) {
        for (int j = 0; j < i; j += 4) {
            out.print(j + " ");
        }
    }
}
```

Question 34

Which of the following is the 8-bit extended two's complement representation of the number to the right?

A) `1111_1101`

B) `1101_1011`

C) `1110_1101`

D) `1101_1110`

E) `1101_1101`

-37_{10}

Question 35

What is output by the code to the right?

A) `9000`

B) `60000`

C) `90000`

D) `120000`

E) There is no output due to an error.

```
enum ChronoUnit {
    MINUTES, HOURS
}

public static void main(String[] args) {
    int ms = 0;
    float time = 1.5f;
    ChronoUnit unit = ChronoUnit.MINUTES;
    switch (unit) {
        case MINUTES:
            int seconds = (int) (time * 60);
            ms = seconds * 1000;
            break;

        case HOURS:
            int minutes = (int) (time * 60);
            int seconds = minutes * 60;
            ms = seconds * 1000;
            break;
    }
    out.println(ms);
}
```

Question 36

What is output by the code to the right?

A) `Monday`

B) `Wednesday`

C) `Sunday`

D) `Invalid day`

E) There is no output due to an error.

```
int day = 3;
String dayName = switch (day) {
    case 1 -> "Monday";
    case 2 -> "Tuesday";
    case 3 -> "Wednesday";
    case 4 -> "Thursday";
    case 5 -> "Friday";
    case 6 -> "Saturday";
    case 7 -> "Sunday";
    default -> "Invalid day";
};
out.println(dayName);
```

Question 37

Which of the following is **not** a possible BFS traversal of the graph to the right? Assume that the BFS began at the first node listed in the answer choice. For example, if the answer is ABC, then the first node visited is A.

A) $ABCDEFG$ **B)** $BDAECGF$

C) $BEADCGF$ **D)** $FCGAEBD$

E) $EBGADFC$

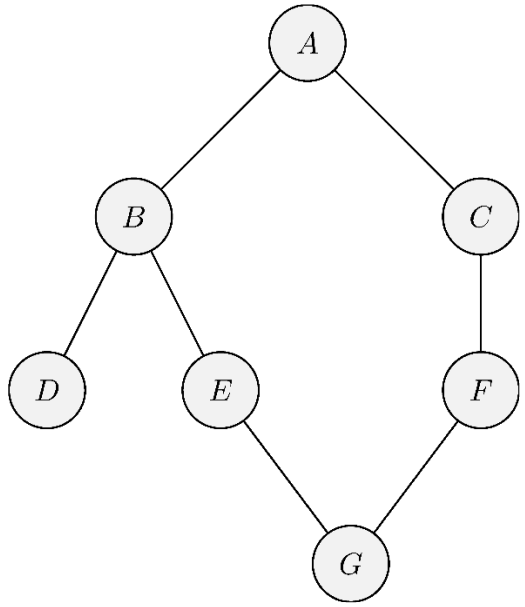
Question 38

Which of the following is **not** a possible DFS traversal of the graph to the right? Assume that the DFS began at the first node listed in the answer choice. For example, if the answer is ABC, then the first node visited is A.

A) $ABDEGCF$ **B)** $DBACFGE$

C) $ABDEGFC$ **D)** $FCABDEG$

E) $EBACFGD$



Question 39 Consider the postfix expression to the right. What is the result when this expression is evaluated? Write your answer in the blank provided on your answer sheet for this question.	`5 3 + 8 2 / * 6 -`
Question 40 How many permutations of `true` and `false` values for the four variables in the expression to the right resolve to `true`? Write your answer in the blank provided on your answer sheet for this question.	$(A \cdot \bar{B} + C) \oplus (\bar{A} + D) \cdot B$